SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether a verifier outside the routed model accepted the result.

PUBLIC-GOODS REQUEST

$150,000

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

808	**168**	**0**	**OPEN**
public GitHub stars	signed prospective local cells	adversarial false passes	MIT code + public proof

Prepared 2026-08-01. GitHub counts are a dated interest signal, not user or installation counts. Application not submitted.

PROBLEM

The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

01 **Declared plan**
Models, tools, topology, limits

02 **Observed execution**
Calls, attempts, artifacts, cost lenses

03 **External verdict**
Deterministic verifier outside the route

04 **Signed evidence**
Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN

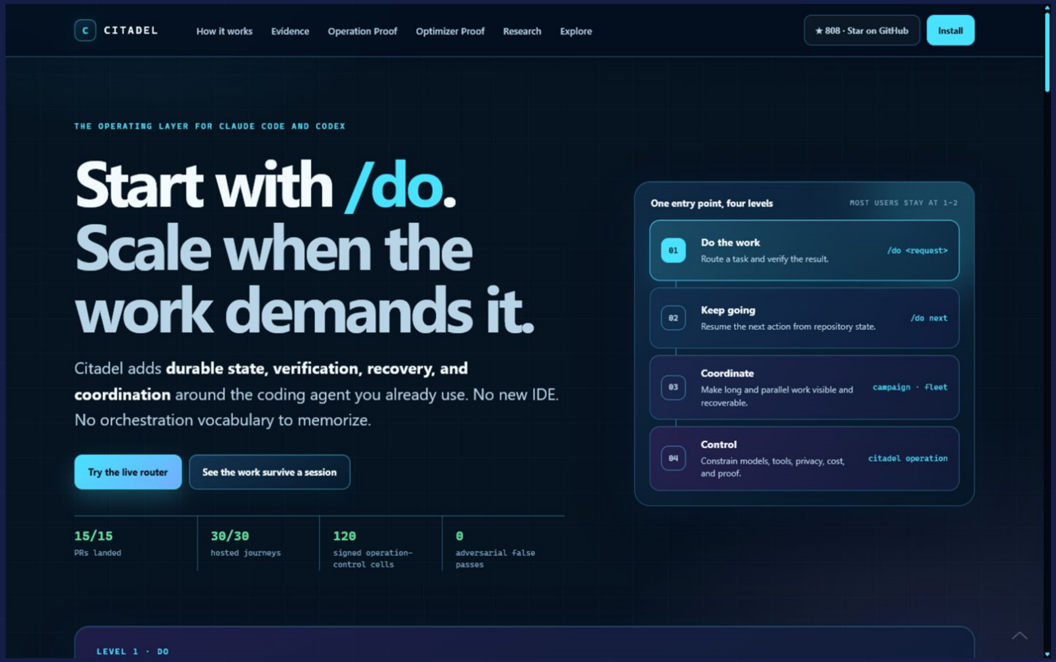
Primary thesis: the unit worth optimizing is a model-externally verified operation, not an isolated model call.

WORKING SUBSTRATE

Built before the grant.

Citadel is usable from a single command, then opens into stronger control only when the operation needs it.

/do	**Beginner path** Natural-language work routing, verification, durable next action.
PLAN	**Portable contract** Models, tools, topology, retries, privacy, time, cost, and stop rules.
RUN	**Runtime coverage** Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
PROOF	**Evidence plane** External artifact verdicts, failure preservation, signed receipts, offline replay.



Public product: https://sethgammon.github.io/Citadel/ | MIT repository: https://github.com/SethGammon/Citadel

PINNED SENTIENT ROMA DIAGNOSTIC

The proof passed. The optimizer gate did not.

A frozen 24-cell run compared frontier, prompt-only routing, direct open/local 7B, and Citadel-controlled ROMA.

POLICY	VERIFIED	DURATION	FRONTIER	LOCAL / MODULE
Frontier-only	6 / 6	47.7s	6	0
Prompt router	3 / 6	40.8s	3	3
Direct open/local 7B	2 / 6	13.2s	0	6
Citadel + ROMA	**4 / 6**	**1042.8s**	**0**	**89 module calls**

24 / 24
measured cells

0
false passes

4 / 6
Citadel local verified

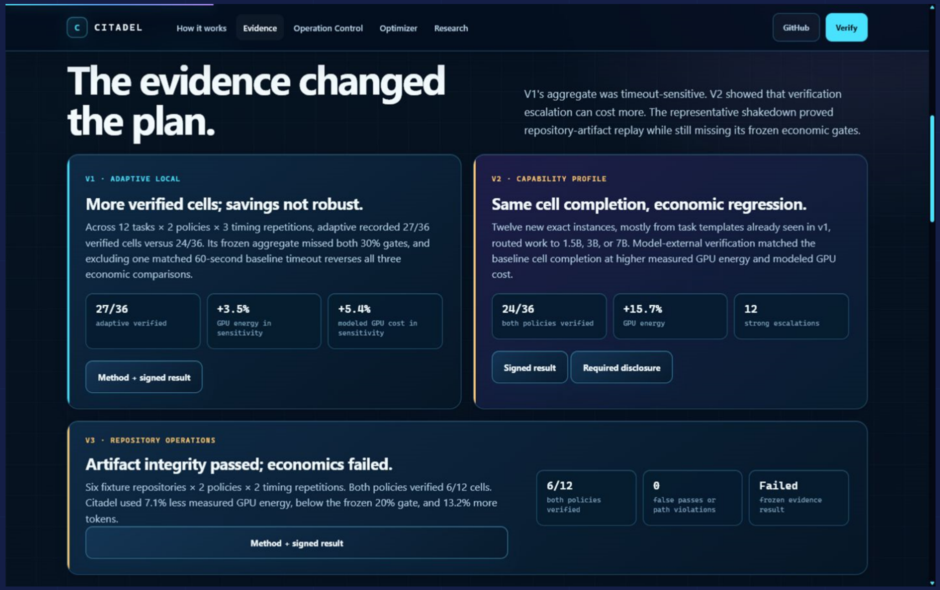
FAILED
strong-attempt savings gate

Evidence: benchmarks/roma-operation-control/published-run/REPORT.md | Offline check: npm run operation-proof:verify

CITADEL

NEGATIVE EVIDENCE IS A DELIVERABLE

Citadel has already rejected its own easy story.



V1 - TIMEOUT SENSITIVITY

27/36 vs 24/36 verified cells. The apparent 9.9% energy saving reversed after excluding one matched timeout pair: +3.5% energy and +5.4% modeled cost.

V2 - CAPABILITY PROFILE

Equal 24/36 completion, but 12 escalations produced +15.7% measured GPU energy and +16.4% modeled cost.

V3 - REPOSITORY ARTIFACTS

Both policies verified 6/12 cells. A 7.1% energy reduction missed the frozen 20% gate; tokens increased 13.2%.

All signed failures remain published under their original identities. Repetitions estimate timing variance, not independent task success.

NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

Funding buys the unproven result.

The grant scales a working control and proof layer into a representative optimizer - or produces a reusable negative result.

01

Representative workload

60+ unique artifact-producing operations; complete cost boundary

$30k

02

Adapter SDK

Sentient ROMA plus two additional open stacks

$28k

03

Expected-value controller

Price verification, escalation, recovery, and failure

$35k

04

Prospective evaluation

Frozen multi-stack policy and statistical contract

$32k

05

Public proof + release

Hosted reconstruction, /do docs, cohort, maintenance

$25k

Canonical milestones and acceptance gates: docs/grants/MILESTONES_AND_BUDGET.md

NO-GO RULES ARE PART OF THE PRODUCT

Clear spend. Hard outcome gates.

COST BASIS

Maintainer labor	$99,000
Compute and tools	$27,000
Hosted reproducibility	$7,000
Docs and accessibility	$7,000
Operator cohort	$2,000
Bounded contingency	$8,000
Total	**$150,000**

PRIMARY PROSPECTIVE GATE

>= 80%
absolute verified completion

>= 95%
of valid frontier rate

>= 30%
lower measured end-to-end cost

ZERO
false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

Tranches: $45k / $60k / $45k. Failed gates terminate in public evidence, failure analysis, reusable artifacts, and unused-funds accounting.

DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

An open optimizer that can prove when it is wrong.

WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- Existing signed evidence lowers substrate risk and exposes the exact remaining research risk.

WHAT STAYS PUBLIC

- MIT code, adapters, conformance contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

Seth Gammon - solo builder and maintainer

OPEN. VERIFIABLE. FALSIFIABLE.

Evaluator path: github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md

Program: https://sentient.foundation/grants | Request: https://sentient.foundation/product-requests | Application packet not submitted.